

The Factor $4/3$

Sphere Volume, Casimir Operators, Electromagnetic Mass, the Triple (3,4,5) and ξ

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The Factor $\frac{4}{3}$

Abstract

The numerical factor $4/3$ in the scaling parameter $\xi = \frac{4}{3} \times 10^{-4}$ has so far been motivated at three places in the corpus: sphere volume (Docs. 006, 205), the Casimir operator of $SU(3)$ and the Casimir effect (Doc. 324). The Casimir effect does not count, because the QFT Casimir energy between ideal plates contains no factor $4/3$. This document adds two occurrences not yet recorded and orders the occurrences by their common cause. First: the factor $4/3$ in the electromagnetic mass of the electron, unexplained since Abraham (1902) and Lorentz (1904), arises because each of the three field components carries one third of the field energy — a consequence of three-dimensionality, not a computational error. Second: the Casimir values of the two smallest non-abelian groups, $C_2(SU(2)) = 3/4$ and $C_2(SU(3)) = 4/3$, are reciprocals of each other; $N = 2$ is the only case in which consecutive fundamental representations achieve this. Both values belong to the electron: $3/4$ is its spin Casimir, $4/3$ its mass factor. The Pythagorean triple $(3, 4, 5)$. The two factors $3/4$ and $4/3$ are reciprocals, and their product is 1. Where does this pair arise naturally? In the simplest right-angled triangle with integer sides: the triple $(3, 4, 5)$. It is not one among many; it is distinguished in four independent ways.

The triple $(3, 4, 5)$ carries this reciprocal pair as $\tan \alpha = 4/3$, $\cot \alpha = 3/4$ and is, by Berggren's theorem, the root of all primitive triples. The angle $\alpha = \arctan(4/3)$ does not occur among the shortest vectors of the lattices D_3 , D_4 , but it does occur on the shell $|v|^2 = 10$: there lie $(3, 1, 0, 0)$ and $(1, 3, 0, 0)$ with $\cos \alpha = 3/5$. The cause is $D_2 = (1 + i)\mathbb{Z}[i]$ and $(1 + i)(2 \pm i) = 1 + 3i$, $3 + i$: the triple sits, via the factor $(1 + i)$, in the carrier of FFGFT. A further 3D occurrence, the enthalpy factor $(u + p)/u = 4/3$ for isotropic pressure $p = u/3$, is added.

Notation

Epistemic markers:

[B]	algebraically proved (Python verification script, exact arithmetic)
[K]	numerically confirmed
[S]	structurally plausible, identification open
[SETTING]	motivated input, not a derivation
[E]	established external mathematics or physics
[X]	checked and negative

Symbol	Meaning
$\xi = 4/30\,000$	dimensionless scaling parameter of FFGFT
$C_2(G)$	quadratic Casimir operator of the fundamental representation of G
$U, \mathbf{p}$	field energy and field momentum of a moving charge
$m_0 = U/c^2$	electromagnetic mass from the rest energy
$(3, 4, 5)$	smallest primitive Pythagorean triple
α	angle with $\tan \alpha = 4/3$; $\alpha \approx 53.13^\circ$
D_3, D_4	root lattices in $\mathbb{R}^3$ (FCC) and $\mathbb{R}^4$
D_2	two-dimensional sublattice; coincides with the ideal $(1 + i) \mathbb{Z}[i]$
$\mathbb{Z}[i]$	ring of Gaussian integers $\{a + bi \mid a, b \in \mathbb{Z}\}$
$(2 + i)$	Gaussian integer with $ 2 + i ^2 = 5$; its square is $(2 + i)^2 = 3 + 4i$
$O(2, 1; \mathbb{Z})$	integer Lorentz group of the form $x^2 + y^2 - z^2$; contains the Berggren matrices
Berggren matrices	three integer 3×3 matrices generating the tree of all primitive triples
ε_0	permittivity of vacuum
e, c	elementary charge, speed of light

1 Starting Point in the Corpus

The parameter $\xi = \frac{4}{3} \times 10^{-4}$ carries two factors. The factor 10^{-4} is treated in Doc. 006 as loop suppression and in Doc. 309 as an exponent setting; it is not the subject of this document. The factor $4/3$ is motivated three times in the corpus:

1. **Sphere volume** (Docs. 006, 205): $V = \frac{4}{3}\pi r^3$; the factor characterises the geometry of three-dimensional space.
2. **Casimir operator** (Doc. 324): $C_2(\text{SU}(3)_{\text{fund}}) = \frac{9-1}{6} = \frac{4}{3}$.
3. **Casimir effect** (Docs. 324, 009): a factor $4/3$ in the Casimir energy between plates.

The third occurrence does not hold: the QFT Casimir energy between ideal plates contains no factor $4/3$, $E/A = -\pi^2 \hbar c / (720 a^3)$, pressure $-\pi^2 \hbar c / (240 a^4)$, energy density $-\pi^2 \hbar c / (720 a^4)$. The factor $4/3$ for the Casimir effect is appended in Doc. 324 (likewise in Doc. 009 and A015); the Casimir effect therefore does not count as an occurrence of $4/3$.

These occurrences stand side by side in the corpus; Doc. 324 calls the agreement of (2) and (3) "no coincidence" without naming a common cause. This document names the common cause of the valid occurrences and adds two occurrences.

2 Electromagnetic Mass: the Factor 4/3 since 1902

The historical finding [E]

If the electron is modelled as a uniformly charged spherical shell of radius a , the field energy at rest is $U = e^2/(8\pi\epsilon_0 a)$ and the associated mass is $m_0 = U/c^2$. If instead one computes the field momentum of the slowly moving shell, one obtains

$$\mathbf{p} = \frac{4}{3} \frac{U}{c^2} \mathbf{v}, \quad (1)$$

i.e. a momentum mass of $\frac{4}{3}m_0$. This factor first appeared in Abraham (1902) and Lorentz (1904). Poincaré (1906) introduced non-electromagnetic binding stresses that remove it computationally; Fermi (1922), Wilson (1936) and Rohrlich (1960) showed that a covariant definition of the four-momentum makes it vanish as well. The literature still disagrees on whether this explains the *cause* or merely redefines the *symptom* (Singal 2024; Universe 11(3):97, 2025).

Why clarification around 1900 was so difficult. The confusion had three mutually reinforcing sources.

First: there was no special relativity. Abraham and Lorentz worked in a framework where energy and momentum were still separate, independent quantities. Without $E = mc^2$ and the four-momentum, a factor 4/3 between momentum and energy looks like a genuine contradiction — not like an artefact of a wrong separation.

Second: the electron was new and its model undecided. Thomson had discovered the electron in 1897. Whether it was a rigid sphere, a deformable shell, or a point mass was an open question. Within Lorentz's sphere model, the factor 4/3 was real and troublesome.

Third: Poincaré's solution was the wrong solution. The "Poincaré pressure" cancelled the factor computationally, but it was an ad-hoc assumption without physical foundation — and it obscured the actual cause.

After Einstein (1905) and Minkowski (1908) the problem could have been resolved cleanly: the factor 4/3 arises only when E and $\mathbf{p}$ are integrated separately from the field, without accounting for the boundary terms of Maxwell's stress tensor. Rohrlich (1960) made this clear. What was never the subject of this debate: that $4/3 = 2(1 - \frac{1}{3})$ is a geometric statement about the three-dimensionality of space. The question asked was "How do we get rid of the factor?" — rather than "Where does it really come from?"

Where the factor comes from [E]

The calculation is short and exposes the cause. With $\mathbf{B} = \mathbf{v} \times \mathbf{E}/c^2$ to first order, the momentum density is $\mathbf{g} = \epsilon_0 \mathbf{E} \times \mathbf{B}$, and

$$\mathbf{p} = \frac{\epsilon_0}{c^2} \int [\mathbf{v} E^2 - \mathbf{E}(\mathbf{E} \cdot \mathbf{v})] dV = \frac{\epsilon_0}{c^2} \mathbf{v} \int (E^2 - E_{\parallel}^2) dV. \quad (2)$$

For a spherically symmetric field, $\langle \cos^2 \theta \rangle = 1/3$, hence $\int E_{\parallel}^2 dV = \frac{1}{3} \int E^2 dV$, and with $U = \frac{\epsilon_0}{2} \int E^2 dV$ it follows that

$$\mathbf{p} = \frac{\epsilon_0}{c^2} \cdot \frac{2}{3} \int E^2 dV \cdot \mathbf{v} = \frac{2}{3} \cdot \frac{2U}{c^2} \mathbf{v} = \frac{4}{3} \frac{U}{c^2} \mathbf{v}. \quad (3)$$

The factor $4/3$ is $2 \cdot (1 - \frac{1}{3})$. It arises because in three-dimensional space each field component carries one third of the energy, and the component parallel to the motion does not contribute to the momentum. This is the same one-third split that sits in the sphere volume $V = \frac{1}{3}rS$ (Archimedes). The factor is not an artefact of a wrong definition but a statement about the dimension of space.

Connection to FFGFT [S]

In Doc. 006 the electron obeys $E_e = 1/\xi_e$ with $\xi_e = \xi \cdot f_e(1, 0, \frac{1}{2}) = \xi = \frac{4}{3} \times 10^{-4}$. The factor $4/3$ thus stands in the numerator of the parameter from which the electron mass is derived (the mass itself is the reciprocal, $1/\xi = (3/4) \cdot 10^4$). Classical electrodynamics has carried, since 1902, the same factor $4/3$ in the electromagnetic electron mass, whose cause is three-dimensionality. FFGFT does not attempt to define it away but sets it as a geometric constant: $\xi = \frac{4}{3} \times 10^{-4}$. This is a structural agreement **[S]**; a derivation of one from the other is not available.

3 Casimir Values: $3/4$ and $4/3$ as a Reciprocal Pair

The calculation [E]

The quadratic Casimir operator measures the total charge of a representation: it is the group-theoretic analogue of $j(j+1)$ for angular momentum. For $SU(2)$ and spin $\frac{1}{2}$ it is $\frac{1}{2}(\frac{1}{2} + 1) = \frac{3}{4}$ — the quantum spin number of the electron. For $SU(3)$ in the fundamental representation (the three colour charges of quarks) it is $\frac{4}{3}$ — the same factor as in the electromagnetic mass.

The quadratic Casimir operator of the fundamental representation of $SU(N)$ is $C_2 = \frac{N^2-1}{2N}$:

Group	C_2	physical location
$SU(2)$	$3/4$	spin $\frac{1}{2}$: $s(s+1) = \frac{3}{4}$
$SU(3)$	$4/3$	colour charge; Doc. 324

We have $C_2(SU(2)) \cdot C_2(SU(3)) = 1$. The verification script shows **[B]**: among all $N \geq 2$, $N = 2$ is the only case in which $C_2(N) \cdot C_2(N+1) = 1$. The two smallest non-abelian groups form a distinguished reciprocal pair.

Both values belong to the electron [S]

The value $3/4$ is the spin Casimir of the electron. The value $4/3$ is, by §2, its electromagnetic mass factor and, in FFGFT, the numerator of $\xi = (4/3) \cdot 10^{-4}$ (the mass reciprocal is $1/\xi = 7500 = (3/4) \cdot 10^4$). Spin and mass of the electron thus carry the reciprocal pair $(3/4, 4/3)$. The fundamental formula $\tilde{T} \cdot m = 1$ is a reciprocal structure; that the two electron invariants also form a reciprocal pair is structurally compatible, but not derived.

4 The Triple (3,4,5)

The reciprocal pair $(3/4, 4/3)$ is the leg ratio of the smallest Pythagorean triple: $\tan \alpha = 4/3$, $\cot \alpha = 3/4$, $\alpha = \arctan \frac{4}{3} \approx 53.13^\circ$, $\sin \alpha = \frac{4}{5}$, $\cos \alpha = \frac{3}{5}$. The triple is not one among many; four properties distinguish it **[E]**, all recomputed in the verification script **[B]**:

1. **Berggren root (1934).** Every primitive Pythagorean triple arises from $(3, 4, 5)$ by a unique sequence of multiplications with three integer 3×3 matrices. The three children of $(3, 4, 5)$ are $(5, 12, 13)$, $(21, 20, 29)$, $(15, 8, 17)$; $(3, 4, 5)$ itself has no positive parent triple. It is the root of the tree.
2. **Arithmetic progression.** $(3, 4, 5)$ is the only right triangle whose sides form an arithmetic progression: $a^2 + (a + d)^2 = (a + 2d)^2 \Leftrightarrow (a - 3d)(a + d) = 0$.
3. **Inradius 1.** $r = (3 + 4 - 5)/2 = 1$. The triples $(5, 12, 13)$ and $(8, 15, 17)$ have $r = 2$ and $r = 3$; $(3, 4, 5)$ is the unit step.
4. **Gaussian integers.** $(2 + i)^2 = 3 + 4i$. The triple is the square of the smallest non-trivial Gaussian integer; α is the angle of this squaring.

The Five in the Triple and Its Connection to the Icosahedral Group [S]

The hypotenuse of the triple is 5, and 5 appears in the trigonometric values: $\sin \alpha = 4/5$, $\cos \alpha = 3/5$. This 5 is not accidentally the same as the 5 in A_5 .

The algebraic chain is as follows. The triple arises as the square of the Gaussian integer $2 + i$: $(2 + i)^2 = 3 + 4i$ [B]. The Gaussian norm is $|2 + i|^2 = 4 + 1 = 5$. The prime 5 splits in $\mathbb{Z}[i]$ as $5 = (2 + i)(2 - i)$ into two conjugate Gaussian prime factors: $N(2 \pm i) = 5$ is prime, so $2 \pm i$ are Gaussian primes and not units; hence 5 is reducible in $\mathbb{Z}[i]$ and not prime there ($5 \equiv 1 \pmod{4}$). The same prime 5 appears in the icosahedral group. The group A_5 has order 60, and the fivefold rotation R_5 has order 5. Its eigenvalue ζ_5 (fifth root of unity) lives in the finite field $\text{GF}(81) = \text{GF}(3^4)$, because $5 \mid 80 = |\text{GF}(81)^*|$ [B]. In the same field $\text{GF}(81)$, via the subfield $\text{GF}(9) \subset \text{GF}(81)$, also lives the golden ratio $\phi = (1 + \sqrt{5})/2$ (Docs. 364, 370).

The connection between the triple $(3, 4, 5)$ and the icosahedral group A_5 thus runs through the prime 5 and the field $\text{GF}(81)$:

$$(2 + i)^2 = 3 + 4i \implies |2 + i|^2 = 5 \implies 5 \mid |\text{GF}(81)^*| = 80 \implies \zeta_5 \in \text{GF}(81).$$

This connects the triple algebraically with the icosahedral part of the corpus (Docs. 285, 293, 367, 370). What this connection is not: a geometric identification. The angle $\alpha = \arctan(4/3) = 53.13^\circ$ is not a rotation angle of A_5 (which has 72° , 144° , 120° , 180°). The connection is algebraic, not geometric [S].

5 Common Cause and Limit

What belongs together [S]

Occurrence	Value	Cause	Source
Sphere volume $V = \frac{1}{3}rS$	$4/3$	one-third split in 3D	Docs. 006, 205
EM mass, field momentum	$4/3$	one-third split in 3D	new, §2
Photon gas, $(u + p)/u$ at $p = u/3$	$4/3$	one-third split in 3D	new, §5
Casimir effect*	$(4/3)$	appended, see below	Doc. 324
$C_2(\text{SU}(3))$	$4/3$	$N = 3$	Doc. 324
$C_2(\text{SU}(2))$	$3/4$	$N = 2$	new, §3
$(3, 4, 5)$	$4/3, 3/4$	Berggren root	new, §4

* The QFT Casimir energy contains no factor $4/3$ (Section 1); the factor is appended in Doc. 324 and does not count as an occurrence.

The enthalpy factor is the calculation of §2 in another language: an isotropic photon gas has pressure $p = u/3$, and the momentum of the moving system is $\mathbf{p} = \frac{4}{3}(E/c^2)\mathbf{v}$ (von Laue)

[E]. The first three rows share a common cause: the number 3 of spatial dimensions and the resulting one-third split. The Casimir rows carry the 3 as $N = 3$ (fundamental representation of $SU(3)$, $C_2 = 4/3$) and as $N^2 - 1 = 3$ (dimension of the adjoint representation of $SU(2)$, $C_2 = 3/4$), not as the rank of the group ($SU(3)$ has rank 2, $SU(2)$ rank 1). That $SU(3)$ and three-dimensional space yield the same factor is, in FFGFT, not a coincidence but follows from the $\mathbb{Z}_3$ triality on D_4 (Doc. 365, layer 4) — this is the point at which the group number and the space number are already identified in the corpus.

The angle α in the D lattice [B]

The D_n lattice is what crystallography calls the face-centred cubic (FCC) lattice: in $\mathbb{R}^n$ all integer vectors whose coordinate sum is even. In FFGFT, D_4 is the carrier lattice of the torus T^4 ; its shortest vectors ($|v|^2 = 2$) define its lattice constant. The question is: on which shells does the angle $\arctan \frac{4}{3} = 53.13^\circ$ appear?

Among the shortest vectors ($|v|^2 = 2$) of the D_3 lattice (FCC) and the D_4 lattice, $\alpha = \arctan \frac{4}{3} = 53.13^\circ$ does not occur; both yield only $60^\circ, 90^\circ, 120^\circ$ **[X]**. On the shell $|v|^2 = 10$ it does occur: $(3, 1, 0, 0)$ and $(1, 3, 0, 0)$ lie in D_4 (even coordinate sum), and $\cos \angle = 6/10 = 3/5$ **[B]**.

The cause is algebraic. The two-dimensional lattice D_2 (even coordinate sum) is exactly the ideal $(1 + i)\mathbb{Z}[i]$ of the Gaussian integers **[B]**, and

$$(1 + i)(2 + i) = 1 + 3i, \quad (1 + i)(2 - i) = 3 + i, \quad (4)$$

with $|1 + 3i|^2 = 10$. The angle between $2 + i$ and $2 - i$ is $\arccos \frac{3}{5} = \alpha$; multiplication by $(1 + i)$ rotates both by 45° and leaves the angle unchanged. The triple $(3, 4, 5)$, which by §4 is the square of $2 + i$, therefore sits, via the factor $(1 + i)$ — the generator of the D lattice — in the carrier geometry of FFGFT, on the first shell whose norm is not composed purely of $|1 + i|^2 = 2$ and $|2|^2 = 4$.

The Berggren matrices are isometries of the quadratic form $x^2 + y^2 - z^2$, i.e. elements of the integer Lorentz group $O(2, 1; \mathbb{Z})$ (Cha and Conceição 2023) **[E]**; the tree of all triples is thus a discrete hyperbolic structure on the same lattice.

The Right Angle in FFGFT — a Further Thread [S]

That the 90° angle plays the key role in §2 has not yet been named explicitly in the text. The derivation shows it: the field component *parallel* to the direction of motion (0°) does not contribute to the field momentum; the two components *perpendicular* to it (90°) do. In three dimensions with three orthogonal directions, one falls out and two remain. That is $2/3$. Multiplied by 2 (from the cross-product structure of $\mathbf{E} \times \mathbf{B}$) gives $4/3$. The geometric statement is:

$$\frac{4}{3} = 2 \cdot \left(1 - \frac{1}{3}\right) = 2 \cdot \frac{\text{number of directions perpendicular to motion}}{3}. \quad (5)$$

The 90° angle is not incidental but the cause. It also appears explicitly in the calculation: $\langle \cos^2 \theta \rangle = 1/3$ follows from spherical symmetry, because this average over all angles is precisely what eliminates the right angle as a mean.

The right angle in the D_4 lattice. Among the shortest vectors of D_4 (even coordinate sum, $|v|^2 = 2$) exactly the angles $60^\circ, 90^\circ, 120^\circ, 180^\circ$ appear **[B]**. The 90° is thus a distinguished lattice angle, but it is not alone: it shares the space of lattice angles with 60° and 120° . Whether the 90° lattice angle plays a distinguished role in FFGFT beyond the others — for instance as the separation angle between the four torus directions, as an orthogonality condition between space and time, or as a condition for the projection $T^4 \rightarrow \mathbb{R}^3 \times S^1$ — has not been investigated. That is the genuinely open question.

The Fourier modes are orthogonal. The three lepton modes v_0, v_1, v_2 form an orthonormal basis: $\langle v_j | v_k \rangle = \delta_{jk}$ **[B]**. Each pair of modes stands at 90° in Hilbert space. This orthogonality is the basic condition for the matrix element $\langle v_0 | R_5 | v_1 \rangle$ to measure the mixing between two *perpendicular* states — and for the modulus squared $|\langle v_0 | R_5 | v_1 \rangle|^2 = 2/9$ to be a transition probability in the sense of Born's rule. Without 90° in Hilbert space, no clean probability interpretation.

2/3 and 4/3 as a bridge to the Koide quotient. The Koide quotient is $Q = (\sum m_k)/(\sum \sqrt{m_k})^2 = 2/3$ **[E]**. The field-momentum formula from §2 gives $\mathbf{p} = \frac{4}{3}(U/c^2)\mathbf{v}$. Numerically, exactly:

$$\frac{4}{3} = 2 \cdot \frac{2}{3} = 2Q.$$

The factor $4/3$ of the electromagnetic mass is twice the Koide quotient. This is a numerical identity — whether a deeper connection lies behind it, between the three-dimensionality factor of field physics and the mass-structure quotient of the leptons, is open **[S]**. It would be the most direct link of the Abraham-Lorentz problem to the lepton masses that the corpus currently contains: via the number $2/3$, which appears on the one hand as the Koide quotient, and on the other as the geometric fraction of transverse directions in three-dimensional space.

The T^4 torus and the four orthogonal circles. The torus T^4 in FFGFT carries four circle directions that in the standard setup stand pairwise at 90° — the simplest metric on the product torus $S^1 \times S^1 \times S^1 \times S^1$. These four 90° pairs generate the D_4 lattice of winding numbers. The transition to spacetime $T^4 \rightarrow \mathbb{R}^3 \times S^1$ picks *one* direction (the mass dimension that becomes time) and leaves the other three as spatial directions, pairwise at 90° . The three-dimensionality of space — that exactly *three* directions can stand perpendicular to a motion — is therefore not a free choice but follows from this projection step. The $4/3$ factor is in this picture the signature of the symmetry that was left behind: the original four-dimensionality of the torus leaves in three-dimensional space exactly the three orthogonal directions that produce $4/3 = 2 \cdot (1 - 1/3)$.

Whether this reading — $4/3$ as a trace of the projection $T^4 \rightarrow \mathbb{R}^3 \times S^1$ — can be formally carried out in the corpus is the sharpest open question of this subsection **[S]**.

What is not shown **[S]**

That the angle occurs in the carrier is shown. That it plays a *role* there — as a mode angle, a winding angle, or in the $\mathbb{Z}_3$ identification — is not shown. The bridge currently reads: the same number $4/3$ at five places with a common cause for three of them, and the associated angle on shell 10 of the carrier lattice. Whether shell 10 is distinguished in FFGFT is open.

6 Result

1. The factor $4/3$ of the electromagnetic electron mass (Abraham 1902) is a three-dimensionality effect: $4/3 = 2(1 - \frac{1}{3})$ **[E][B]**. It is the numerator of $\xi = (4/3) \cdot 10^{-4}$, not of $1/\xi = 7500 = (3/4) \cdot 10^4$ **[S]**.
2. $C_2(\text{SU}(2)) = 3/4$ and $C_2(\text{SU}(3)) = 4/3$ are reciprocals; $N = 2$ is the only such case **[B]**. Both values are invariants of the electron (spin, mass) **[S]**.
3. $(3, 4, 5)$ is the Berggren root, the only triple in arithmetic progression, has inradius 1, and is the square of $2 + i$ **[B]**. It carries the reciprocal pair as its legs.

4. The angle $\arctan \frac{4}{3}$ is absent among the shortest vectors of D_3, D_4 **[X]**, but occurs on the shell $|v|^2 = 10$ because $D_2 = (1+i)\mathbb{Z}[i]$ and $(1+i)(2 \pm i) = 1 + 3i, 3 + i$ **[B]**. A role of this shell in FFGFT is open **[S]**.
5. The enthalpy factor $(u+p)/u = 4/3$ at $p = u/3$ is a further occurrence of the one-third split **[E]**.

Verification script: 2/python/Dok371_Skripte/ffgft_371_faktor_vier_drittel.py — 28 checks, 28 passed (P1–P5, P7 positive; P6: negative control of the shortest vectors and positive finding on shell 10).

Register (Doc. 190): no entry — no earlier document is corrected; the occurrences in Docs. 006, 205, 324 remain valid and are supplemented.

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